

Experiment No. 9

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Experiment Title

Implementation of Web Storage using localStorage and sessionStorage; Development of a User Preferences System and Seminar Schedule Planner

Software / Tools Required

- Visual Studio Code
- Google Chrome
- HTML5
- JavaScript (ES6)
- Browser Web Storage API

Practical — User Preferences

Aim

To develop a User Preferences web application using JavaScript Web Storage, where theme and accent color preferences are stored using localStorage and the username is stored using sessionStorage.

File Path

<file:///Users/ojaswinithote/Desktop/Javascript/EXP%209/Practical/index.html>

file:///Users/ojaswinithote/Desktop/Javascript/EXP%209/Case%20Study/index1.html

Program Code

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Preferences</title>

  <style>
    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    :root {
      --background: #f4f4f4;
      --card: #ffffff;
      --text: #222222;
      --secondary: #666666;
      --border: #dddddd;
      --accent: #7658f;
    }
  </style>
</head>
<body>
```

```

body {
  font-family: Arial, sans-serif;
  background: var(--background);
  color: var(--text);
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  transition: 0.3s ease;
}

body.dark {
  --background: #181818;
  --card: #252525;
  --text: #f5f5f5;
  --secondary: #bdbdbd;
  --border: #444444;
}

.container {
  width: 90%;
  max-width: 550px;
  background: var(--card);
  padding: 35px;
  border-radius: 18px;
  box-shadow: 0 10px 30px rgba(0, 0, 0, 0.15);
  transition: 0.3s ease;
}

h1 {
  text-align: center;
  color: var(--accent);
  margin-bottom: 10px;
}

.description {
  text-align: center;
  color: var(--secondary);
  margin-bottom: 30px;
}

.setting {
  margin-bottom: 22px;
}

label {
  display: block;
  font-weight: bold;
  margin-bottom: 8px;
}

select,
input {
  width: 100%;
  padding: 12px;
  border: 2px solid var(--border);
  border-radius: 8px;
  background: var(--card);
  color: var(--text);
  font-size: 15px;
  outline: none;
}

select:focus,
input:focus {
  border-color: var(--accent);
}

```

```

button {
  width: 100%;
  padding: 13px;
  border: none;
  border-radius: 8px;
  background: var(--accent);
  color: white;
  font-size: 16px;
  cursor: pointer;
  margin-top: 8px;
  transition: 0.2s;
}

button:hover {
  opacity: 0.85;
}

#message {
  text-align: center;
  margin-top: 18px;
  color: var(--accent);
  font-weight: bold;
  min-height: 20px;
}

.storage-info {
  margin-top: 25px;
  padding: 15px;
  background: var(--background);
  border-radius: 10px;
  font-size: 14px;
  line-height: 1.7;
  color: var(--secondary);
}

.storage-info strong {
  color: var(--accent);
}
</style>
</head>

<body>

  <div class="container">

    <h1>User Preferences</h1>

    <p class="description">
      Customize and save your personal preferences.
    </p>

    <div class="setting">
      <label for="username">Username</label>
      <input type="text" id="username" placeholder="Enter your name">
    </div>

    <div class="setting">
      <label for="theme">Choose Theme</label>

      <select id="theme">
        <option value="light">Light</option>
        <option value="dark">Dark</option>
      </select>
    </div>

    <div class="setting">
      <label for="accent">Choose Accent Color</label>

      <select id="accent">

```

```

        <option value="purple">Purple</option>
        <option value="blue">Blue</option>
        <option value="green">Green</option>
    </select>
</div>

<button id="saveButton">Save Preferences</button>

<button id="clearButton">Clear Preferences</button>

<div id="message"></div>

<div class="storage-info">
    <strong>localStorage:</strong>
    Theme and accent color are saved permanently.
    <br>
    <strong>sessionStorage:</strong>
    Username is saved for the current session.
</div>

</div>

<script src="script.js"></script>

</body>
</html>

```

script.js

```

const username = document.getElementById("username");
const theme = document.getElementById("theme");
const accent = document.getElementById("accent");
const saveButton = document.getElementById("saveButton");
const clearButton = document.getElementById("clearButton");
const message = document.getElementById("message");

function applyTheme(selectedTheme) {
    if (selectedTheme === "dark") {
        document.body.classList.add("dark");
    } else {
        document.body.classList.remove("dark");
    }
}

function applyAccent(selectedAccent) {
    if (selectedAccent === "blue") {
        document.documentElement.style.setProperty("--accent", "#3f6fa8");
    } else if (selectedAccent === "green") {
        document.documentElement.style.setProperty("--accent", "#4f805b");
    } else {
        document.documentElement.style.setProperty("--accent", "#76558f");
    }
}

const savedTheme = localStorage.getItem("theme");
const savedAccent = localStorage.getItem("accent");
const savedUsername = sessionStorage.getItem("username");

if (savedTheme) {
    theme.value = savedTheme;
    applyTheme(savedTheme);
}

if (savedAccent) {
    accent.value = savedAccent;
    applyAccent(savedAccent);
}

if (savedUsername) {

```

```

    username.value = savedUsername;
}

theme.addEventListener("change", function () {
    applyTheme(theme.value);
});

accent.addEventListener("change", function () {
    applyAccent(accent.value);
});

username.addEventListener("input", function () {
    message.textContent = "";
});

saveButton.addEventListener("click", function () {
    localStorage.setItem("theme", theme.value);
    localStorage.setItem("accent", accent.value);

    sessionStorage.setItem("username", username.value);

    applyTheme(theme.value);
    applyAccent(accent.value);

    message.textContent = "Preferences saved successfully!";
});

clearButton.addEventListener("click", function () {
    localStorage.removeItem("theme");
    localStorage.removeItem("accent");

    sessionStorage.removeItem("username");

    theme.value = "light";
    accent.value = "purple";
    username.value = "";

    applyTheme("light");
    applyAccent("purple");

    message.textContent = "Preferences cleared!";
});

```

Features

- Enter a username.
- Choose Light or Dark theme.
- Choose Purple, Blue, or Green accent color.
- Save theme and accent color using localStorage.
- Save username using sessionStorage.
- Save or clear preferences.

Storage Used

Storage	Data Stored	Persistence
localStorage	Theme and accent color	Persists after browser restart.
sessionStorage	Username	Current browser session.

JavaScript Concepts / Functions Used

Concept / Function	Purpose
localStorage.setItem()	Stores theme and accent color.
localStorage.getItem()	Retrieves saved preferences.

<code>localStorage.removeItem()</code>	Removes saved local preferences.
<code>sessionStorage.setItem()</code>	Stores username for the session.
<code>sessionStorage.getItem()</code>	Retrieves the username.
<code>sessionStorage.removeItem()</code>	Removes the username.
<code>addEventListener()</code>	Handles Save and Clear button events.
DOM Manipulation	Updates theme, accent color, and messages.

Output

- User Preferences interface is displayed.
- Theme can be changed between Light and Dark.
- Accent color can be selected.
- Theme and accent color are stored in `localStorage`.
- Username is stored in `sessionStorage`.
- Stored preferences can be cleared.

Screenshot

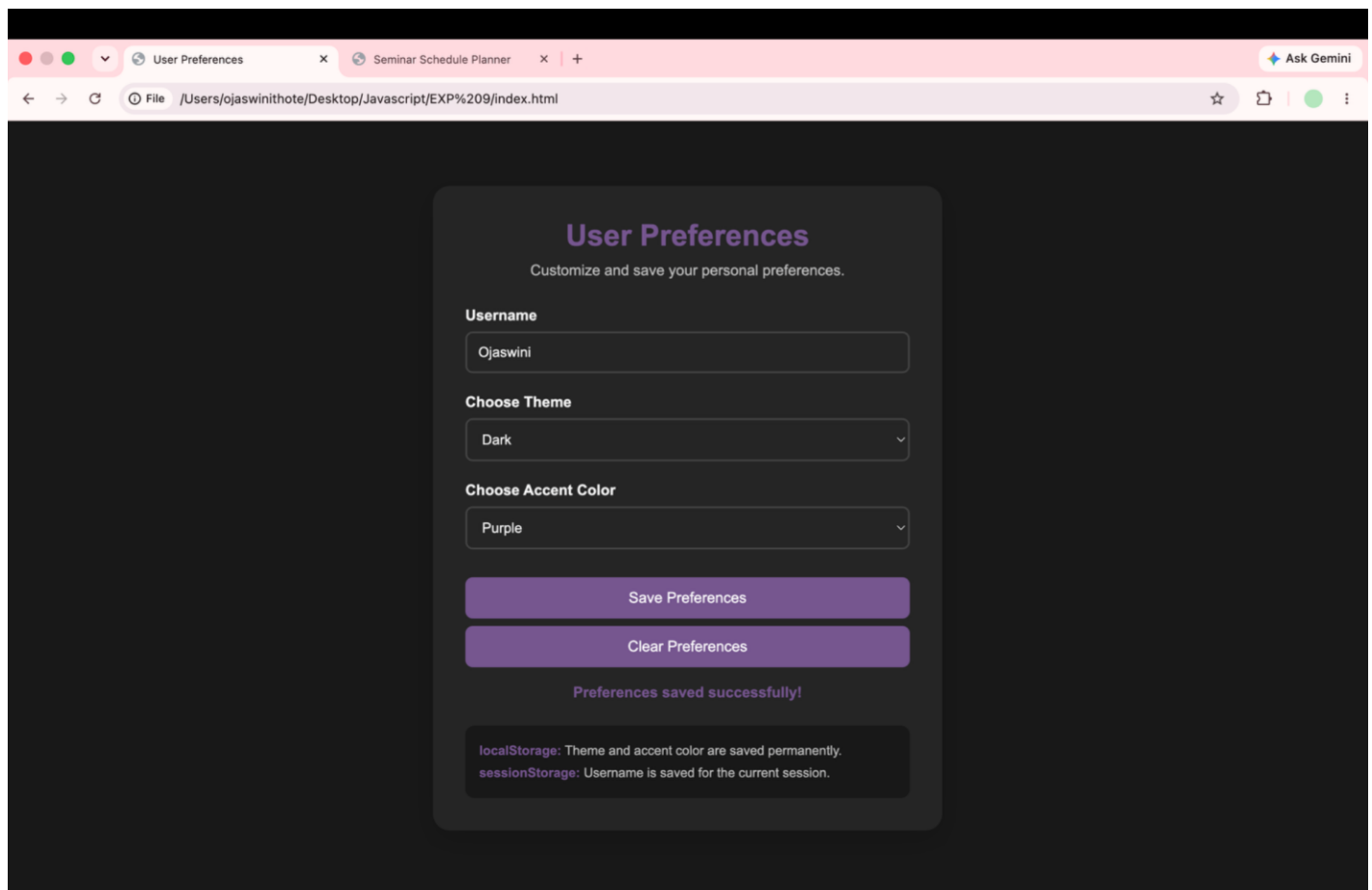


Figure 1: User Preferences — Practical Output

Case Study — Seminar Schedule Planner

Aim

To develop an interactive Seminar Schedule Planner that uses JavaScript event handling and Web Storage to save a selected seminar topic in localStorage and the selected seminar day in sessionStorage.

File Path

Case Study/index1.html

Case Study/script1.js

Program Code

index1.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Seminar Schedule Planner</title>

  <style>
    * {
      box-sizing: border-box;
    }

    body {
      margin: 0;
      padding: 40px 20px;
      font-family: Arial, sans-serif;
      background-color: #f5f0fa;
      color: #333;
    }

    .container {
      max-width: 900px;
      margin: auto;
      background-color: white;
      padding: 30px;
      border-radius: 15px;
      box-shadow: 0 5px 20px rgba(80, 60, 100, 0.15);
    }

    h1 {
      text-align: center;
      color: #654c7c;
      margin-bottom: 10px;
    }

    .subtitle {
      text-align: center;
      color: #777;
      margin-bottom: 30px;
    }

    table {
      width: 100%;
      border-collapse: collapse;
      margin-bottom: 25px;
    }

    th {
```

```

        background-color: #d9c7e8;
        color: #4c385d;
        padding: 14px;
        border: 1px solid #bca8cc;
    }

    td {
        padding: 14px;
        text-align: center;
        border: 1px solid #d4c8dc;
    }

    tbody tr {
        cursor: pointer;
        transition: 0.2s;
    }

    tbody tr:hover {
        background-color: #f1eafa;
    }

    tbody tr.selected {
        background-color: #dcd0ee;
        font-weight: bold;
    }

    .day {
        font-weight: bold;
        color: #654c7c;
    }

    .time {
        color: #555;
    }

    .topic {
        font-weight: 500;
    }

    .buttons {
        display: flex;
        justify-content: center;
        gap: 12px;
        flex-wrap: wrap;
        margin-top: 20px;
    }

    button {
        padding: 12px 18px;
        border: none;
        border-radius: 8px;
        background-color: #76588f;
        color: white;
        font-size: 14px;
        cursor: pointer;
    }

    button:hover {
        background-color: #5e4672;
    }

    #clearButton {
        background-color: #9a6b6b;
    }

    #clearButton:hover {
        background-color: #7f5555;
    }

```

```

#status {
  text-align: center;
  margin-top: 20px;
  padding: 12px;
  color: #654c7c;
  font-weight: bold;
}

.storage-info {
  margin-top: 20px;
  padding: 18px;
  background-color: #f8f5fb;
  border-radius: 10px;
  border: 1px solid #ddd2e5;
}

.storage-info p {
  margin: 10px 0;
}

.storage-info strong {
  color: #654c7c;
}

@media (max-width: 650px) {
  body {
    padding: 20px 10px;
  }

  .container {
    padding: 15px;
  }

  table {
    font-size: 13px;
  }

  th,
  td {
    padding: 9px 5px;
  }

  button {
    width: 100%;
  }
}
</style>
</head>

<body>

  <div class="container">

    <h1>Seminar Schedule Planner</h1>
    <p class="subtitle">
      Click on a seminar to view its details
    </p>

    <table>
      <thead>
        <tr>
          <th rowspan="2">Day</th>
          <th colspan="2">Schedule</th>
          <th rowspan="2">Topic</th>
        </tr>
        <tr>
          <th>Begin</th>
          <th>End</th>
        </tr>
      </thead>
    </table>
  </div>

```

```

</thead>

<tbody>
  <tr data-day="Monday" data-topic="Introduction to XML">
    <td class="day" rowspan="2">Monday</td>
    <td class="time">8:00 a.m.</td>
    <td class="time">5:00 p.m.</td>
    <td class="topic">Introduction to XML</td>
  </tr>

  <tr data-day="Monday" data-topic="Validity: DTD and Relax NG">
    <td class="time">8:00 a.m.</td>
    <td class="time">11:00 a.m.</td>
    <td class="topic">Validity: DTD and Relax NG</td>
  </tr>

  <tr data-day="Tuesday" data-topic="XPath">
    <td class="day" rowspan="2">Tuesday</td>
    <td class="time">11:00 a.m.</td>
    <td class="time">2:00 p.m.</td>
    <td class="topic">XPath</td>
  </tr>

  <tr data-day="Tuesday" data-topic="XSL Transformations">
    <td class="time">2:00 p.m.</td>
    <td class="time">5:00 p.m.</td>
    <td class="topic">XSL Transformations</td>
  </tr>

  <tr data-day="Wednesday" data-topic="XSL Formatting Objects">
    <td class="day">Wednesday</td>
    <td class="time">8:00 a.m.</td>
    <td class="time">12:00 p.m.</td>
    <td class="topic">XSL Formatting Objects</td>
  </tr>
</tbody>
</table>

<div class="buttons">
  <button id="saveTopicButton">
    Save Topic (localStorage)
  </button>

  <button id="saveDayButton">
    Save Day (sessionStorage)
  </button>

  <button id="clearButton">
    Clear All
  </button>
</div>

<div id="status">
  Select a seminar to begin.
</div>

<div class="storage-info">
  <p>
    <strong>localStorage:</strong>
    <span id="localData">Not saved</span>
  </p>

  <p>
    <strong>sessionStorage:</strong>
    <span id="sessionData">Not saved</span>
  </p>
</div>

</div>

```

```

<script src="script1.js"></script>

</body>
</html>

```

script1.js

```

const rows = document.querySelectorAll("tbody tr");

const saveTopicButton = document.getElementById("saveTopicButton");
const saveDayButton = document.getElementById("saveDayButton");
const clearButton = document.getElementById("clearButton");

const status = document.getElementById("status");
const localData = document.getElementById("localData");
const sessionData = document.getElementById("sessionData");

let selectedTopic = "";
let selectedDay = "";

const savedTopic = localStorage.getItem("selectedTopic");
const savedDay = sessionStorage.getItem("selectedDay");

if (savedTopic) {
    selectedTopic = savedTopic;
    localData.textContent = savedTopic;

    rows.forEach(function (row) {
        if (row.getAttribute("data-topic") === savedTopic) {
            row.classList.add("selected");
        }
    });

    status.textContent = "Saved Topic: " + savedTopic;
}

if (savedDay) {
    selectedDay = savedDay;
    sessionData.textContent = savedDay;
}

rows.forEach(function (row) {
    row.addEventListener("click", function () {
        selectedTopic = row.getAttribute("data-topic");
        selectedDay = row.getAttribute("data-day");

        const cells = row.querySelectorAll("td");

        let beginTime;
        let endTime;

        if (cells.length === 4) {
            beginTime = cells[1].textContent;
            endTime = cells[2].textContent;
        } else {
            beginTime = cells[0].textContent;
            endTime = cells[1].textContent;
        }

        rows.forEach(function (item) {
            item.classList.remove("selected");
        });

        row.classList.add("selected");

        alert(
            "Seminar Details\n\n" +
            "Topic: " + selectedTopic +

```

```

        "\nDay: " + selectedDay +
        "\nTime: " + beginTime + " - " + endTime
    );

    status.textContent =
        "Selected: " + selectedTopic + " (" + selectedDay + ")";
    });
});

saveTopicButton.addEventListener("click", function () {
    if (selectedTopic === "") {
        alert("Please select a seminar first.");
        return;
    }

    localStorage.setItem("selectedTopic", selectedTopic);

    localData.textContent =
        localStorage.getItem("selectedTopic");

    status.textContent =
        "Topic saved in localStorage: " + selectedTopic;

    alert("Topic saved in localStorage!");
});

saveDayButton.addEventListener("click", function () {
    if (selectedDay === "") {
        alert("Please select a seminar first.");
        return;
    }

    sessionStorage.setItem("selectedDay", selectedDay);

    sessionData.textContent =
        sessionStorage.getItem("selectedDay");

    status.textContent =
        "Day saved in sessionStorage: " + selectedDay;

    alert("Day saved in sessionStorage!");
});

clearButton.addEventListener("click", function () {
    localStorage.removeItem("selectedTopic");
    sessionStorage.removeItem("selectedDay");

    selectedTopic = "";
    selectedDay = "";

    rows.forEach(function (row) {
        row.classList.remove("selected");
    });

    localData.textContent = "Not saved";
    sessionData.textContent = "Not saved";

    status.textContent = "All saved preferences cleared.";

    alert("localStorage and sessionStorage cleared!");
});

```

Features

- Displays seminar schedules in a table.
- Allows a seminar row to be selected.
- Displays selected seminar details.

- Saves selected topic using localStorage.
- Saves selected day using sessionStorage.
- Displays stored values.
- Provides Clear All functionality.
- Uses a responsive layout.

Seminar Schedule

Day	Begin	End	Topic
Monday	8:00 a.m.	5:00 p.m.	Introduction to XML
Monday	8:00 a.m.	11:00 a.m.	Validity: DTD and Relax NG
Tuesday	11:00 a.m.	2:00 p.m.	XPath
Tuesday	2:00 p.m.	5:00 p.m.	XSL Transformations
Wednesday	8:00 a.m.	12:00 p.m.	XSL Formatting Objects

JavaScript Concepts / Functions Used

Concept / Function	Purpose
DOM Manipulation	Accesses seminar rows and updates page information.
addEventListener()	Handles seminar and button clicks.
localStorage	Stores the selected topic.
sessionStorage	Stores the selected day.
setItem()	Saves selected data.
getItem()	Retrieves stored data.
removeItem() / clear()	Removes stored data.
dataset	Reads day and topic values from rows.

Output

- Seminars are displayed by day and time.
- Clicking a seminar selects and highlights it.
- Selected topic can be saved to localStorage.
- Selected day can be saved to sessionStorage.
- Saved values are displayed in the storage section.
- Clear All removes saved data.

Screenshot

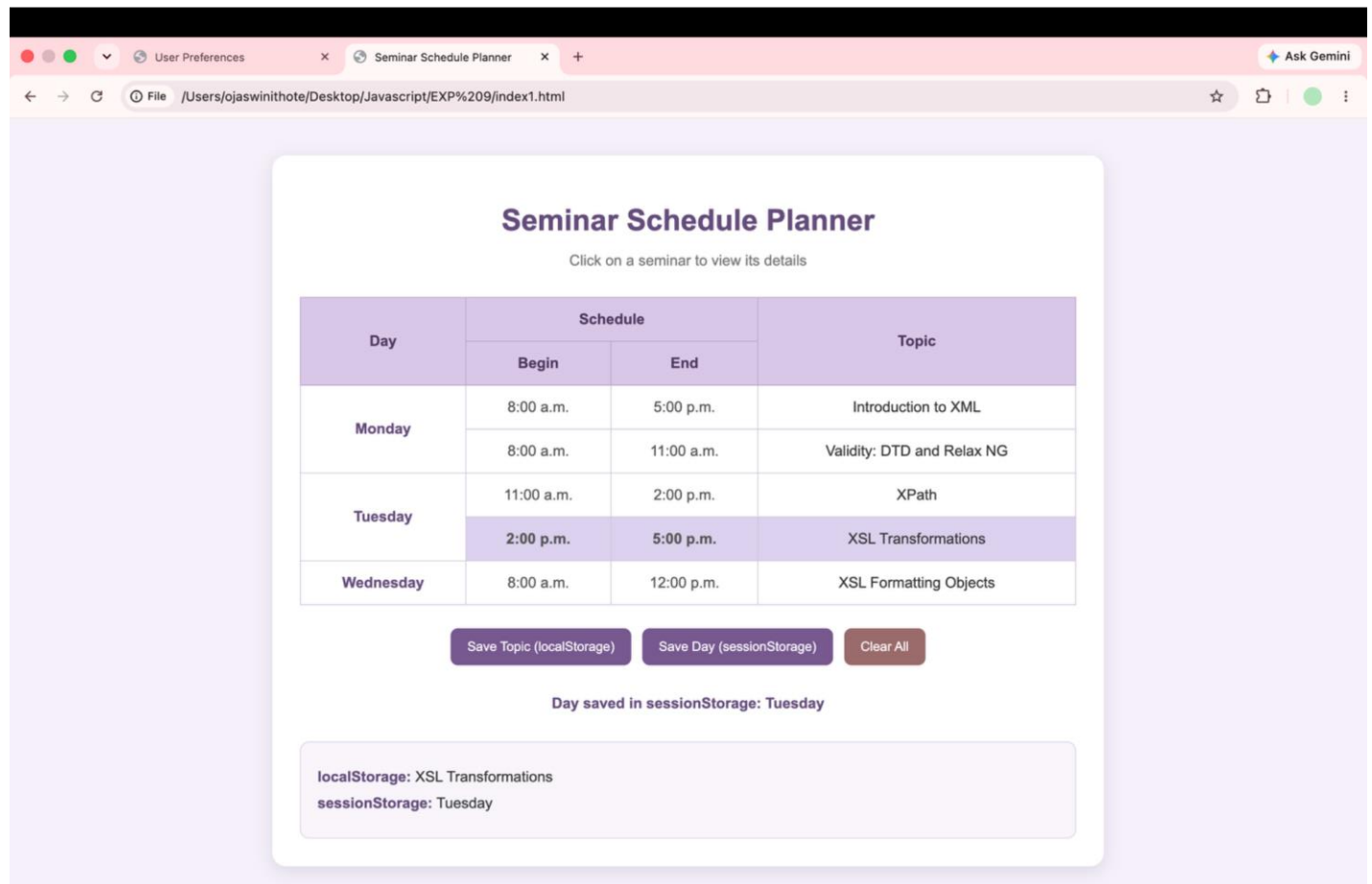


Figure 2: Seminar Schedule Planner — Case Study Output

Result / Conclusion

Both the Practical and Case Study were completed successfully. The Practical demonstrates `localStorage` and `sessionStorage` for saving user preferences and managing browser data persistence. The Case Study applies the same Web Storage concepts to an interactive Seminar Schedule Planner, where the selected topic is stored using `localStorage` and the selected day is stored using `sessionStorage`. Thus, the experiment demonstrates the practical use of the Web Storage API, DOM manipulation, and event handling in JavaScript.